

From Local Response to Global Gravitational Charges: Asymptotic Momentum and Stationary Boundaries in Motion–Staticity Geometry

Mingyang Han

School of Mathematical Sciences, Shanghai Jiao Tong University

Shanghai 200240, China

hanmingyang@sjtu.edu.cn

October 6, 2026

Abstract

We apply the classical motion–staticity (MS) response construction to gravitational charges at spatial infinity and at a stationary horizon. The local pairing $\tilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi)$ supplies the calibrated soldered metric; the torsion-free Einstein action and its coupling are separate dynamical inputs. On a Regge–Teitelboim phase space, the standard differentiability boundary term defines a total translation covector $\mathcal{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$. Its scalar observer energy reconstructs the corresponding Lorentz vector, and its minimum gives the invariant mass when that vector is future timelike. We calculate which asymptotic response-scale jets enter the translation surface functionals: the monopole of the leading spatial scale and the dipole of its normal first derivative. These are kinematic readouts; the constraints restrict the realizable jets, as a conformal initial-data example illustrates. A Hermitian representation connects the global compact variables to the trace-one section of the positive Lorentz–Hermitian cone. Using the same Einstein normalization, we then compare the particle proper-time source, the ADM Hamiltonians and the stationary horizon boost charge. The Hamiltonian completion, observer-space geometry, conformal method and black-hole identities retain their established status. The contribution is their explicit realization and comparison in calibrated MS response variables, rather than a new charge formula or a derivation of Newton’s constant.

Keywords: motion–staticity geometry; asymptotic flatness; Hamiltonian boundary terms; ADM energy–momentum; covariant phase space; stationary horizons; Lorentz–Hermitian cone

Contents

1	Introduction	2
2	Operational response polarity	4
2.1	Local massive-response polarity	4
2.2	Causal closure, projective boundary and calibration	5
3	Calibrated asymptotically Minkowskian data	5
3.1	First-jet response calibration	6
3.2	Asymptotic translations	6
4	Boundary completion and the total translation covector	7
4.1	Differentiability obstruction	7
4.2	Translation-dual charge and vacuum normalization	8

5	Translation symmetry, covariance and conservation	8
5.1	Reduced translation algebra	8
5.2	Asymptotic Lorentz covariance	8
6	Observer-space geometry of total momentum	9
6.1	Causal classification and local–global reciprocity	10
6.2	Hermitian interface for a future-causal total charge	10
7	Response calibration, constraints and ADM identification	10
7.1	Translation-visible response jets	11
7.2	Constraint compatibility of pure response rescalings	11
7.3	Physical units and ADM realization	12
8	Matter sources and stationary boundary charges	13
8.1	The proper-time response as a matter source	13
8.2	One covariant boundary variation	14
8.3	Horizon boost charge and the first law	14
8.4	Komar normalization and the remaining coupling	15
9	Discussion and outlook	15
A	Finite-radius boundary variation	16
B	Asymptotic translation estimates and conventions	17
C	Observer-space reconstruction and calibration quotients	17
C.1	Intrinsic characterization of linear observer energies	17
C.2	Translation-visible calibration quotient	18

1 Introduction

A local generally covariant theory does not by itself assign a total energy or momentum to an isolated gravitating system. In canonical general relativity, the Hamiltonian and momentum constraints generate hypersurface deformations, but nontrivial translation charges appear only after one specifies asymptotic structure and completes the bulk generators so that they are functionally differentiable. For asymptotically flat data this is the Regge–Teitelboim construction, whose translation-sector boundary terms are the ADM energy–momentum [1–3].

There are close precedents for treating gravitational energy as a response of an action to boundary data. Brown and York obtain a boundary stress tensor by varying the gravitational action with respect to the boundary metric [4]. Covariant Hamiltonian formulations also relate reference geometry, displacement fields, boundary conditions and gravitational charges, including their black-hole applications [5]. Those constructions do not use the primitive MS pairing, but they already establish the general boundary-response and common variational framework. The present paper applies that framework to the geometry reconstructed from MS; it does not claim priority for the framework itself.

The paper has two related aims. First, we ask where the asymptotic translation charge enters when the local spacetime geometry and the compact-support bulk generator have already been obtained from the motion–staticity (MS) response construction. Second, once that translation sector is fixed, we ask how the same metric calibration and Einstein normalization appear in

the standard covariant boundary variation at a stationary inner boundary. Neither question requires a new ADM or black-hole-mechanics formula; the issue is whether these familiar charges fit coherently into one response-derived geometry.

The flat MS theory starts from the compact state (D, S) satisfying $|D|^2 + S^2 = 1$ with $S > 0$. Its projective image may be written $\Phi(D, S) = S^{-1}(e_0 + D^i e_i)$ in a chosen future-oriented orthonormal frame. The free-particle analysis of Ref. [6] selects the usual positive massive shell and the Lorentzian observer geometry. The curved-spacetime reconstruction of Ref. [7] keeps the internal Lorentz generator space distinct from $T_x M$ and reconstructs the missing relation from the scalar response

$$\tilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi), \quad e_x : T_x M \xrightarrow{\sim} V_x, \quad g = e^* \eta. \quad (1)$$

The nondegeneracy and reconstruction hypotheses of the preceding work are retained here. The field e is the usual soldering/coframe field once it has been reconstructed. Selecting the torsion-free Einstein branch supplies the standard hypersurface-deformation generator; the scalar pairing alone does not select that dynamics. The quantum configuration-space branch developed in Ref. [8] is not used in the present paper.

The passage from local geometry to a total momentum is not obtained by adding local particle momenta. In a generic curved spacetime there is no preferred global Abelian translation group, and gravitational binding prevents the total mass from being a sum of local rest scales. The natural global object is instead dual to the asymptotic remnant of translations and is determined by the Hamiltonian boundary problem.

We therefore consider a spacelike hypersurface with one asymptotically Minkowskian end, calibrated relative to a flat reference response. Constant leading lapse and shift define a four-dimensional quotient translation space $\mathfrak{t}_\infty \simeq \mathbb{R} \oplus \mathbb{R}^3$. At this stage $\mathfrak{t}_\infty$ contains only translation parameters. Given $a \in \mathfrak{t}_\infty$, the reconstructed compact-support generator extends to a noncompact smearing as a finite bulk functional, but its variation contains a boundary obstruction. The differentiable generator has the form

$$G[a] = G_0[\chi_a] + B_\infty[a], \quad B_\infty[a] = a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}}. \quad (2)$$

Linearity in a then defines the total translation covector $\mathcal{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$ by $\mathcal{P}_\infty^{\text{MS}}(a) = B_\infty[a]$. This is the first point at which a global momentum-like object appears in the construction.

Once the charge has been produced, its Lorentzian interpretation is independent of the details of the boundary calculation. If P_∞^{MS} denotes the Lorentz-polar vector and u a future unit asymptotic observer, the scalar energy $E_\infty(u) = -\eta_\infty(P_\infty^{\text{MS}}, u)$ obeys

$$P_\infty^{\text{MS}} = E_\infty u - \nabla_H E_\infty, \quad -\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}}) = E_\infty^2 - |\nabla_H E_\infty|^2. \quad (3)$$

When the vector is future timelike, the invariant radius is the minimum of E_∞ on the future observer hyperboloid. This global minimization is the counterpart of the local response polarity, where a displacement is fixed and the massive generator is varied.

The asymptotic response calibration and the gravitational normalization play different roles. A positive recalibration $\Theta \mapsto \lambda \Theta$ changes the reconstructed geometry by $g \mapsto \lambda^2 g$ and can alter the boundary data. At the translation-supporting orders, a leading spatial scale $\lambda - 1 = \sigma(\hat{x})/r + \dots$ changes the energy only through the monopole of σ , while a normal first jet $n(\ln \lambda) = \rho(\hat{x})/r^2 + \dots$ changes the spatial translation coefficient only through the dipole of ρ . These statements concern the surface functional. The constraints impose additional restrictions, and in the pure-rescaling family the leading normal jet is in fact ruled out by the stronger off-shell constraint decay used below.

Only after the charge has been constructed do we impose the physical Einstein normalization $\kappa = 8\pi G/c^4$. With the sign convention for extrinsic curvature stated below, the completed

energy and physical momentum are then the standard ADM quantities. This identification does not determine G ; it fixes how the already reconstructed geometry is assigned a gravitational charge.

Positive energy is logically later still. The observer-space classification applies to an arbitrary completed translation vector and therefore includes spacelike, null and past-directed possibilities. After the ADM identification, the standard positive-mass theorem may be invoked under its own completeness, decay and dominant-energy hypotheses [9–11]. A separate companion note examines null-response formulations of that positivity problem and their relation to Sen–Witten frames [12]; no such theorem is used in the construction below.

The same distinction between geometry and dynamics is important at a horizon. Once the Einstein action and its boundary prescription are adopted, the covariant phase-space variation supplies Hamiltonians at infinity and Noether charges at an inner stationary boundary. We use this standard framework only after the ADM construction is complete. The stationary section therefore records a compatibility of local response calibration, matter coupling and boundary charges; it does not derive black-hole mechanics from the primitive response.

The scope of the contribution is consequently specific: we express the standard translation charge in response-derived variables, identify its sensitivity to response calibration, and compare the local and global observer descriptions. The response-jet formulas below are calculated explicitly, but no claim of historical priority is made for the underlying conformal, Hamiltonian or Lorentzian identities.

The paper is organized as follows. [Section 2](#) records the local response polarity needed later. [Section 3](#) defines the calibrated asymptotically flat sector and the quotient translation space. [Section 4](#) constructs the differentiable generator and total translation covector. [Section 5](#) treats the reduced translation algebra, asymptotic Lorentz covariance and conservation. [Section 6](#) develops the observer-energy representation and gives a brief Hermitian interface for a future-causal total charge. [Section 7](#) separates response calibration, constraint compatibility and gravitational normalization, and then identifies the charge with the ADM four-momentum. [Section 8](#) places matter sources, asymptotic Hamiltonians and the stationary horizon boost charge in one covariant variational framework. [Section 9](#) summarizes the resulting classical boundary architecture.

2 Operational response polarity

We use signature $(-, +, +, +)$. Whenever units are displayed, a positive-energy momentum vector and its canonical action covector are written

$$P^a = \left(\frac{E}{c}, \mathbf{p} \right), \quad \Pi_a = P_a^b = \left(-\frac{E}{c}, \mathbf{p} \right). \quad (4)$$

Except in [Section 7.3](#), we set $c = 1$.

2.1 Local massive-response polarity

Let V_x be the internal Lorentz space at x and let

$$\mathcal{H}_{Q,x}^+ := \{P \in V_x : \eta_x(P, P) = -Q^2, P \text{ future directed}\}, \quad Q > 0. \quad (5)$$

For $P \in \mathcal{H}_{Q,x}^+$ set $\Pi = P^b$ and $\hat{\Pi} = -\Pi = -P^b$. The sign-reversed covector is convenient because it gives a positive response on future-timelike displacements. Define

$$\Theta_x^+(\hat{\Pi}, \xi) := \hat{\Pi}(e_x \xi), \quad \mu_x(\xi) := \frac{1}{Q} \inf_{\hat{\Pi} \in \hat{\Sigma}_{Q,x}^+} \Theta_x^+(\hat{\Pi}, \xi), \quad (6)$$

where $\hat{\Sigma}_{Q,x}^+ = \{-P^b : P \in \mathcal{H}_{Q,x}^+\}$.

Proposition 2.1 (Local response polarity). *For every future-timelike $\xi \in T_x M$,*

$$\mu_x(\xi) = \sqrt{-g_x(\xi, \xi)}. \quad (7)$$

The infimum is attained at a unique covector, whose pullback to $T_x^ M$ is*

$$\hat{p}_\xi = -\frac{Q}{\mu_x(\xi)} g_x^\flat(\xi) = Q d\mu_x|_\xi. \quad (8)$$

Moreover, the metric is recovered from the same scalar functional by

$$g_x(v, w) = -\frac{1}{2} D^2(\mu_x^2)_\xi(v, w). \quad (9)$$

Proof. Write $Y = e_x \xi$ and $P = QU$ with $\eta(U, U) = -1$. Then $Q^{-1}\Theta_x^+ = -\eta(U, Y)$. The reverse Lorentzian Cauchy–Schwarz inequality gives $-\eta(U, Y) \geq \sqrt{-\eta(Y, Y)}$, with equality exactly when $U = Y/\sqrt{-\eta(Y, Y)}$. Since $g = e^*\eta$, this proves (7); differentiating $\mu_x^2 = -g_x(\xi, \xi)$ gives the remaining statements. $\square$

The Lorentzian inequality and its equality case are standard [13]; the point here is not that inequality, but the provenance of the dual shell: it is the massive generator sector already selected in the preceding MS construction rather than an independently postulated family of covectors.

2.2 Causal closure, projective boundary and calibration

The same infimum may be extended to arbitrary $\xi \in T_x M$. Its value is $\sqrt{-g_x(\xi, \xi)}$ on the future-timelike cone, vanishes on the nonzero future-null cone and at $\xi = 0$, and is unbounded below for spacelike or nonzero past-directed vectors. For a future-null ξ , zero is an infimum but not a minimum; a minimizing sequence requires unbounded rapidity and approaches the projective $S \rightarrow 0$ boundary of the compact MS state.

This observation separates two operations that will remain distinct at infinity. An internal Lorentz gauge transformation $e_x \mapsto \Lambda_x e_x$ leaves the response scalar, g_x and μ_x unchanged when the generator transforms dually. By contrast, a positive recalibration $\tilde{\Theta}_x \mapsto \lambda(x)\tilde{\Theta}_x$ rescales $e_x \mapsto \lambda e_x$, hence $g_x \mapsto \lambda^2 g_x$ and $\mu_x \mapsto \lambda \mu_x$. The causal cone is projective and survives the rescaling, but proper-time and metric normalization do not.

This distinction is the local precursor of the asymptotic calibration problem. The observer-independent causal directions can be discussed before an absolute scale is fixed; a dimensionful total translation charge cannot.

3 Calibrated asymptotically Minkowskian data

Let Σ have one end $\Sigma_\infty \simeq \mathbb{R}^3 \setminus B_R$ with asymptotically Cartesian coordinates x^i , $r = (\delta_{ij} x^i x^j)^{1/2}$ and $\hat{x}^i = x^i/r$. The calibrated reference response determines a Minkowski metric whose reference slice has $\bar{h}_{ij} = \delta_{ij}$ and $\bar{K}_{ij} = 0$. We write $\bar{D}$ for the Euclidean connection and use $O_k(r^{-p})$ to mean that up to k reference derivatives have the corresponding differentiated falloff.

The physical response reconstructs a spacetime metric g , with induced data (h_{ij}, K_{ij}) on Σ . We adopt

$$K_{ij} = \frac{1}{2N} (\dot{h}_{ij} - \mathcal{L}_\beta h_{ij}), \quad \pi^{ij} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij}), \quad (10)$$

so π^{ij} is a symmetric contravariant density of weight one. This convention is the negative of a common ADM convention for K_{ij} ; the conversion is made explicitly in Section 7.3.

Definition 3.1 (Calibrated asymptotically Minkowskian MS end). A configuration belongs to $\Gamma_\infty^{\text{MS}}$ if, in the calibrated end,

$$h_{ij} = \delta_{ij} + \frac{a_{ij}(\hat{x})}{r} + O_2(r^{-2}), \quad \pi^{ij} = \frac{b^{ij}(\hat{x})}{r^2} + O_1(r^{-3}), \quad (11)$$

with Regge–Teitelboim parity $a_{ij}(-\hat{x}) = a_{ij}(\hat{x})$ and $b^{ij}(-\hat{x}) = -b^{ij}(\hat{x})$. Admissible tangent variations preserve these conditions and the calibration and constraint-decay conditions below.

The usual parity argument makes the canonical symplectic form finite: the potentially logarithmic term is proportional to $\int dr r^{-1} \int_{\mathbb{S}^2} b^{ij} \delta a_{ij} d\Omega$, whose angular integral vanishes by parity [1, 3]. Equation (10) equivalently gives $K_{ij} = k_{ij}(\hat{x})r^{-2} + O_1(r^{-3})$ with odd k_{ij} .

3.1 First-jet response calibration

A recalibration $\Theta \mapsto \lambda\Theta$ rescales $g \mapsto \lambda^2 g$. If

$$\lambda = 1 + \frac{\sigma(\hat{x})}{r} + o(r^{-1}), \quad n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + o(r^{-2}), \quad (12)$$

where the scale remainder has two controlled spatial derivatives and the normal-jet remainder has one, then the leading metric coefficient changes by $a_{ij} \mapsto a_{ij} + 2\sigma\delta_{ij}$, while the normal first jet changes the r^{-2} boost data. Preserving the Regge–Teitelboim parity class requires σ even and ρ odd. A convenient sufficient calibration condition is therefore

$$\lambda = 1 + o(r^{-1}), \quad \bar{D}_i \lambda = o(r^{-2}), \quad n(\ln \lambda) = o(r^{-2}). \quad (13)$$

This condition removes residual recalibration at the orders capable of changing the four translation surface functionals. It does not set the physical r^{-1} metric or r^{-2} momentum coefficients to zero; those coefficients label different calibrated configurations.

For the off-shell differentiability problem we also impose the sufficient decay

$$\mathcal{H}_\perp = O(r^{-3-\epsilon}), \quad \mathcal{H}_i = O(r^{-3-\epsilon}), \quad \epsilon > 0, \quad (14)$$

with the same decay for their admissible tangent variations. On the physical constraint surface, $\mathcal{H}_\perp \approx 0$ and $\mathcal{H}_i \approx 0$. We assume throughout that the Hamiltonian flows called admissible preserve this phase space, including its falloff, parity, calibration and constraint-decay conditions. These are sufficient domain assumptions, not a claim that the domain is minimal.

3.2 Asymptotic translations

An admissible translation representative has

$$N = a^\perp + O_1(r^{-1}), \quad \beta^i = a^i + O_1(r^{-1}), \quad (15)$$

where $(a^\perp, a^i)$ is constant relative to the calibrated reference end. Representatives differing by a lapse–shift pair that vanishes at infinity define the same asymptotic translation. The quotient is denoted $\mathfrak{t}_\infty \simeq \mathbb{R} \oplus \mathbb{R}^3$.

Remark 3.2 (Asymptotic Hamiltonian admissibility). A deformation approaching $a \in \mathfrak{t}_\infty$ represents a Hamiltonian asymptotic translation on $\Gamma_\infty^{\text{MS}}$ only if its generator is finite, integrable in field space and functionally differentiable.

This is the usual Hamiltonian differentiability requirement [1], not an independent MS postulate or a second derivation of the Einstein constraints. It tests whether the compact-support generator already present in the classical MS branch admits a nontrivial asymptotic extension.

4 Boundary completion and the total translation covector

With $c = 1$, the bulk constraints are

$$\mathcal{H}_\perp = \frac{2\kappa}{\sqrt{h}} \left(\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} {}^{(3)}R, \quad \mathcal{H}_i = -2D_j \pi^j_i, \quad (16)$$

and the bare generator is $G_0[\chi] = \int_\Sigma (N\mathcal{H}_\perp + \beta^i \mathcal{H}_i) d^3x$. The decay (14) makes this finite for (15), but finiteness does not remove the boundary term in its variation.

4.1 Differentiability obstruction

Let Σ_R be the truncation of the chosen outer end and S_R its external sphere. Other ends or inner boundaries are held fixed or assigned smearings and boundary conditions for which their surface variations vanish; otherwise they require separate boundary terms. The stationary horizon in section 8 is treated separately. The lapse and shift descriptors are held fixed in the variations below. The derivative boundary term from the scalar-curvature part of the normal generator is

$$\mathcal{S}_R^\perp[N; \delta h] = -\frac{1}{2\kappa} \oint_{S_R} s^i \left[N(D^j \delta h_{ij} - D_i \delta h) - (D^j N) \delta h_{ij} + (D_i N) \delta h \right] dS_h. \quad (17)$$

Under the asymptotic conditions above, only the constant part of N survives. Writing $q_{ij} = h_{ij} - \delta_{ij}$, the corresponding primitive is

$$E_\infty^{\text{MS}} := \frac{1}{2\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j_j) \bar{s}^i dS_{\bar{h}}. \quad (18)$$

Thus $\mathcal{S}_R^\perp = -a^\perp \delta E_\infty^{\text{MS}} + o(1)$.

For the momentum constraint, integration by parts is most transparent in density notation. If $dS_j = \frac{1}{2} \epsilon_{jkl} dx^k \wedge dx^l$, then

$$-2 \int_{\Sigma_R} \beta^i D_j \pi^j_i d^3x = \int_{\Sigma_R} \pi^{ij} \mathcal{L}_\beta h_{ij} d^3x - 2 \oint_{S_R} \beta^i \pi_i^j dS_j. \quad (19)$$

The finite-radius metric remainder in the variation decays as $O(R^{-1})$, and the subleading shift contributes at the same order. The surviving boundary coefficient is therefore

$$J_i^{\text{MS}} := 2 \lim_{R \rightarrow \infty} \oint_{S_R} \pi_i^j dS_j = \frac{1}{\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j_i - K \delta^j_i) s_j dS_h. \quad (20)$$

Theorem 4.1 (Regge–Teitelboim completion in response variables). *For $a = (a^\perp, a^i) \in \mathfrak{t}_\infty$, define*

$$B_\infty[a] = a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}}. \quad (21)$$

Then the bare generator $G_0[\chi_a]$ is finite but generally not differentiable for nonzero a . Its surface obstruction is $-\delta B_\infty[a]$, and

$$G[a] = G_0[\chi_a] + B_\infty[a] \quad (22)$$

is finite and functionally differentiable on $\Gamma_\infty^{\text{MS}}$.

Proof. The kinetic term in $\mathcal{H}_\perp$ is ultralocal and has no derivative boundary contribution. Equation (17) reduces asymptotically to $-a^\perp \delta E_\infty^{\text{MS}}$. In the tangential sector, (19) leaves $-a^i \delta J_i^{\text{MS}}$; all terms involving subleading smearings or the finite-radius metric remainder vanish in the limit. Their sum is $-\delta B_\infty[a]$. $\square$

4.2 Translation-dual charge and vacuum normalization

Differentiability fixes $B_\infty[a]$ up to a field-independent constant on each connected component of phase space. The calibrated Minkowski configuration $h_{ij} = \delta_{ij}$, $\pi^{ij} = 0$ fixes the origin by $B_\infty[a] = 0$ there.

Definition 4.2 (Total asymptotic translation covector). The total MS translation covector is the element $\mathcal{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$ defined by

$$\mathcal{P}_\infty^{\text{MS}}(a) = B_\infty[a] = a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}}. \quad (23)$$

On a component not containing the calibrated vacuum, the same explicit surface prescription still defines the charge, but differentiability alone does not transport the vacuum normalization across disconnected components.

The construction is deliberately translation-sector only. Boost and rotation generators require their own asymptotic smearings, parity analysis and boundary completion [1, 3]; none is needed to define (23).

5 Translation symmetry, covariance and conservation

5.1 Reduced translation algebra

Choose for each $a \in \mathfrak{t}_\infty$ a representative $\chi_a = (N_a, \beta_a)$ that equals a constant translation outside a compact set. The surface-deformation bracket of two such representatives has vanishing constant part, so it is proper: $[\chi_a, \chi_b]_{\text{SD}} = \chi_{ab}^{\text{proper}}$. Hence the quotient algebra is Abelian.

The charges themselves have no residual translation-sector bracket. An asymptotic translation changes h_{ij} only by $O(r^{-2})$ and π^{ij} by $O(r^{-3})$, so the leading coefficients that enter (18) and (20) are invariant. Therefore

Theorem 5.1 (Reduced translation algebra). *Assume that the translation flows preserve $\Gamma_\infty^{\text{MS}}$. On the physical constraint surface, after quotienting proper deformations,*

$$\{\mathcal{P}_\infty^{\text{MS}}(a), \mathcal{P}_\infty^{\text{MS}}(b)\}_{\text{red}} = 0, \quad a, b \in \mathfrak{t}_\infty. \quad (24)$$

No global statement about possible central extensions of the full hypersurface-deformation algebra is required for this translation result.

5.2 Asymptotic Lorentz covariance

The calibrated reference end carries a Minkowski bilinear form η_∞ on $\mathfrak{t}_\infty$. If two admissible asymptotic inertial descriptions are related by $\Lambda_\infty \in SO^+(1, 3)$, then $a \mapsto a' = \Lambda_\infty a$. Since the Hamiltonian pairing is representation-independent,

$$\mathcal{P}_\infty^{\text{MS}'}(a') = \mathcal{P}_\infty^{\text{MS}}(a), \quad \mathcal{P}_\infty^{\text{MS}'} = (\Lambda_\infty^{-1})^* \mathcal{P}_\infty^{\text{MS}}. \quad (25)$$

We use $\mathcal{P}_\infty^{\text{MS}}$ for the energy-valued Hamiltonian covector and P_∞^{MS} for its Lorentz-polar vector; they are distinct objects. In the present $c = 1$ convention the associated vector is

$$P_\infty^{\text{MS}} := -\eta_\infty^\#(\mathcal{P}_\infty^{\text{MS}}), \quad (26)$$

which transforms as $P_\infty^{\text{MS}'} = \Lambda_\infty P_\infty^{\text{MS}}$.

Equation (25) is the covariance of the abstract asymptotic translation covector. Interpreting it as a comparison of charges computed on different asymptotic hyperplanes uses the standard

geometric covariance of ADM momentum and is not merely a change of basis in one fixed Hamiltonian chart [3, 14].

If the physical evolution is generated by a differentiable Hamiltonian whose asymptotic class is a fixed $t \in \mathfrak{t}_\infty$, and its flow preserves $\Gamma_\infty^{\text{MS}}$, then [theorem 5.1](#) gives $\frac{d}{dt}\mathcal{P}_\infty^{\text{MS}}(a) = \{\mathcal{P}_\infty^{\text{MS}}(a), \mathcal{P}_\infty^{\text{MS}}(t)\}_{\text{red}} = 0$ for every fixed a . Thus both the covector and its Lorentz-polar vector are conserved on the reduced physical phase space.

6 Observer-space geometry of total momentum

The translation charge has been constructed without assuming an invariant mass. We now describe its scalar readout by asymptotic observers. The identities in this section are standard properties of ambient linear functions and hyperbolic static potentials [15–17]; their role is to connect the completed boundary charge to the same compact observer variables that occur in the local MS theory.

Let

$$H_\infty^+ := \{u \in \mathfrak{t}_\infty : \eta_\infty(u, u) = -1, \text{ } u \text{ future directed}\} \quad (27)$$

with induced Riemannian metric h_H . Define the observer energy by $E_\infty(u) = \mathcal{P}_\infty^{\text{MS}}(u) = -\eta_\infty(P_\infty^{\text{MS}}, u)$. In a future-oriented orthonormal basis, write $P_\infty^{\text{MS}} = E_\infty^{\text{MS}}e_0 + p_\infty^i e_i$ (still with $c = 1$). Parameterizing u by MS variables gives

$$u(D, S) = \frac{1}{S}(e_0 + D^i e_i), \quad E_\infty(D, S) = \frac{E_\infty^{\text{MS}} - p_\infty \cdot D}{S}, \quad |D|^2 + S^2 = 1, \quad S > 0. \quad (28)$$

Proposition 6.1 (Momentum reconstruction from the observer-energy field). *At every $u \in H_\infty^+$,*

$$\nabla_H E_\infty = -P_\infty^{\text{MS}} + E_\infty u, \quad P_\infty^{\text{MS}} = E_\infty u - \nabla_H E_\infty. \quad (29)$$

Consequently,

$$-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}}) = E_\infty^2 - |\nabla_H E_\infty|_{h_H}^2, \quad \text{Hess}_H E_\infty = E_\infty h_H. \quad (30)$$

In $3 + 1$ dimensions, $\Delta_H E_\infty = 3E_\infty$.

Proof. For $v \in T_u H_\infty^+$, $\eta_\infty(u, v) = 0$. The vector $-P_\infty^{\text{MS}} + E_\infty u$ is tangent and pairs with v as $-\eta_\infty(P_\infty^{\text{MS}}, v) = dE_\infty(v)$. This proves (29). Taking the Lorentzian norm gives the first identity in (30); differentiating tangentially gives the Hessian equation. $\square$

If P_∞^{MS} is future timelike, set $Q_\infty^{\text{MS}} = \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})}$ and $u_* = P_\infty^{\text{MS}}/Q_\infty^{\text{MS}}$. Then

$$E_\infty(u) = Q_\infty^{\text{MS}} \cosh d_H(u, u_*), \quad Q_\infty^{\text{MS}} = \min_{H_\infty^+} E_\infty, \quad (31)$$

with a unique minimum at u_* . At that observer $\nabla_H E_\infty = 0$ and $\text{Hess}_H E_\infty = Q_\infty^{\text{MS}} h_H$: the invariant radius is both the minimum energy and the isotropic Hessian scale there. Moreover, $\text{Hess}_H E_\infty \geq Q_\infty^{\text{MS}} h_H$, so the energy is strictly geodesically convex. These are consequences of the standard hyperboloid identities, not additional mass laws. Observer minimization also appears in other gravitational mass settings, for example the asymptotically anti-de Sitter construction of Chen, Hung, Wang and Yau [18]; their AdS charge and observer class are not used here. In compact coordinates, $D_\infty = p_\infty/E_\infty^{\text{MS}}$ and $S_\infty = Q_\infty^{\text{MS}}/E_\infty^{\text{MS}}$, so $P_\infty^{\text{MS}} = Q_\infty^{\text{MS}} \Phi(D_\infty, S_\infty)$.

6.1 Causal classification and local–global reciprocity

The boundary construction itself does not imply that P_∞^{MS} is future timelike. The energy landscape gives the complete classification:

$$\inf_{u \in H_\infty^+} E_\infty(u) = \begin{cases} Q_\infty^{\text{MS}} > 0, & P_\infty^{\text{MS}} \text{ future timelike,} \\ 0, & P_\infty^{\text{MS}} \text{ future null or zero,} \\ -\infty, & P_\infty^{\text{MS}} \text{ spacelike or nonzero past directed.} \end{cases} \quad (32)$$

For a nonzero future-null vector, the infimum is approached only as $S \rightarrow 0^+$ in the aligned infinite-rapidity direction. Thus P_∞^{MS} is future causal exactly when $E_\infty(u) \geq 0$ for every future unit observer. This is Lorentz-cone duality, not a positive-mass theorem.

There is nevertheless a useful structural relation to [section 2](#). Locally, a displacement ξ is fixed and the positive massive generator is varied to obtain $\mu_x(\xi) = \sqrt{-g_x(\xi, \xi)}$. Globally, after boundary completion, the total generator is fixed and the observer is varied to obtain Q_∞^{MS} in the future-timelike sector. The two minima use the same Lorentzian polarity with the roles of probe and generator exchanged. No relation such as $Q_\infty^{\text{MS}} = \sum_A Q_A$ follows; the global radial invariant is a boundary functional of the full gravitational data.

6.2 Hermitian interface for a future-causal total charge

There is a useful algebraic way to display the same compact observer variables without changing the Hamiltonian construction. Choose an oriented asymptotic spin frame and let σ_i be the Pauli matrices. The total Lorentz vector of [\(26\)](#) may be represented by the Hermitian matrix

$$\mathbf{P}_\infty := E_\infty^{\text{MS}} I - p_\infty^i \sigma_i, \quad \det \mathbf{P}_\infty = (E_\infty^{\text{MS}})^2 - |p_\infty|^2. \quad (33)$$

This is the standard vector–Hermitian-matrix correspondence between the future Lorentz cone and the positive semidefinite 2×2 Hermitian cone [\[19\]](#); no spinor field on Σ is being introduced.

If the completed vector is future causal and $E_\infty^{\text{MS}} > 0$, then $\mathbf{P}_\infty$ is positive semidefinite. Its normalized representative

$$\varrho_\infty := \frac{\mathbf{P}_\infty}{2E_\infty^{\text{MS}}} = \frac{1}{2}(I - D_\infty^i \sigma_i), \quad D_\infty := \frac{p_\infty}{E_\infty^{\text{MS}}}, \quad (34)$$

therefore belongs to the closed trace-one convex section of the positive Hermitian cone, equivalently the Bloch ball. In the future-timelike sector, $S_\infty := Q_\infty^{\text{MS}}/E_\infty^{\text{MS}}$ obeys $|D_\infty|^2 + S_\infty^2 = 1$ and $\det \varrho_\infty = S_\infty^2/4$. Thus the observer coordinates of [\(28\)](#) are determinant coordinates on that normalized section. The resemblance to a qubit density matrix is algebraic: no physical density operator, Hilbert-space representation or quantum boundary Hamiltonian is constructed here.

This representation is not used to prove future causality. Before a positive-mass theorem is supplied, imposing $\varrho_\infty \geq 0$ would simply assume the desired conclusion. The rank-one boundary is the usual pure-state/null-direction boundary of this correspondence [\[19\]](#); null-response positivity in MS variables is examined separately in Ref. [\[12\]](#).

7 Response calibration, constraints and ADM identification

The construction contains two scale choices with different meanings. Response calibration fixes the asymptotic geometry because $\Theta \mapsto \lambda\Theta$ changes g . The coefficient κ , by contrast, fixes the gravitational charge assigned to a fixed canonical geometry. We now separate the two effects and restore physical units.

7.1 Translation-visible response jets

Temporarily relax (13) to the leading jets (12). The metric change is $\Delta h_{ij} = 2\sigma\delta_{ij}/r + o(r^{-1})$, and direct substitution in (18) gives

$$\Delta_\lambda E_\infty^{\text{MS}} = \frac{2}{\kappa} \int_{\mathbb{S}^2} \sigma \, d\Omega. \quad (35)$$

Thus only the monopole of the even function σ is visible to the temporal translation charge.

For the normal first jet, the conformal transformation of the extrinsic curvature gives $K'^j_i - K'\delta^j_i = \lambda^{-1}[K^j_i - K\delta^j_i - 2n(\ln \lambda)\delta^j_i]$. Hence

$$\Delta_\lambda J_i^{\text{MS}} = -\frac{2}{\kappa} \int_{\mathbb{S}^2} \rho(\hat{x}) \hat{x}_i \, d\Omega, \quad (36)$$

so only the $\ell = 1$ part of the odd function ρ contributes. The surface-functional readout is therefore the finite projection

$$(\sigma, \rho) \mapsto \left(\int_{\mathbb{S}^2} \sigma \, d\Omega, \int_{\mathbb{S}^2} \rho \hat{x}_i \, d\Omega \right) \in \mathbb{R} \oplus \mathbb{R}^3. \quad (37)$$

The temporal shift is the standard conformal ADM-mass formula, written with spatial factor $h \mapsto \lambda^2 h$ rather than $h \mapsto \phi^4 h$; setting $\phi = \sqrt{\lambda}$ gives the usual mass transformation [20]. The vector projection follows from the conformal extrinsic-curvature law and spherical-harmonic orthogonality. Their role here is to identify which tails of the primitive response enter the already known translation surface functional. This statement concerns that functional before imposing the constraints. Higher even modes of σ and higher odd modes of ρ are invisible to the four translations, although they may matter for other asymptotic observables.

At fixed geometric data (h, K) , a rescaling $\kappa \mapsto \alpha\kappa$ with $\alpha > 0$ entails $\pi^{ij} \mapsto \alpha^{-1}\pi^{ij}$ and gives $(E_\infty^{\text{MS}}, p_\infty, Q_\infty^{\text{MS}}) \mapsto \alpha^{-1}(E_\infty^{\text{MS}}, p_\infty, Q_\infty^{\text{MS}})$ while leaving (D_∞, S_∞) unchanged. This comparison changes the relative normalization of the gravitational theory, rather than providing a gauge transformation in a theory with fixed coupling. Response recalibration changes the geometry; gravitational normalization changes the radial charge assigned to that geometry.

7.2 Constraint compatibility of pure response rescalings

The response jets in (37) are not freely specifiable physical modes. The stronger off-shell constraint decay (14) already removes most of the pure-rescaling family.

Proposition 7.1 (Constraint restriction on pure response rescaling). *Assume the original data satisfy (14). Let the full spacetime metric be rescaled by $g' = \lambda^2 g$ with the differentiated asymptotics (12). If the transformed data satisfy the same constraint decay, then*

$$\Delta_{\mathbb{S}^2} \sigma = 0, \quad \rho = 0. \quad (38)$$

Thus smooth σ is constant, and a nonzero normal scale jet cannot by itself generate an admissible spatial-momentum change.

Proof. Write $v = n(\ln \lambda)$. The conformal transformation of (10) gives $\pi'^{ij} = \pi^{ij} - (\sqrt{h}/\kappa)h^{ij}v$. Using the three-dimensional scalar-curvature transformation and the asymptotic falloff, the leading changes in the constraints are

$$\mathcal{H}'_\perp - \mathcal{H}_\perp = \frac{2}{\kappa r^3} \Delta_{\mathbb{S}^2} \sigma + o(r^{-3}), \quad \mathcal{H}'_i - \mathcal{H}_i = \frac{2}{\kappa} \partial_i \left(\frac{\rho}{r^2} \right) + o(r^{-3}). \quad (39)$$

The required $O(r^{-3-\epsilon})$ decay removes both displayed r^{-3} terms. A smooth harmonic function on $\mathbb{S}^2$ is constant; contracting the second expression with $\hat{x}^i$ gives $-4\rho/(\kappa r^3) + o(r^{-3})$, hence $\rho = 0$. $\square$

The scalar restriction is consistent with the curvature-integrability requirements in conformal mass comparisons [20]. The normal-jet restriction follows from the leading momentum constraint under the specified decay. These are necessary asymptotic conditions, not a global existence theorem. In particular, the momentum of an admissible data set must be carried by additional normal deformation data, such as a trace-free part of K_{ij} .

A standard conformal initial-data example makes the distinction concrete. We use the Lichnerowicz–York conformal method [21, 22] with a Bowen–York momentum seed. Take $h_{ij} = \psi^4 \delta_{ij}$, $K = 0$ and $\pi^{ij} = \psi^{-4} \tilde{\pi}^{ij}$. A Bowen–York translation seed with target physical momentum p_∞^i in the present $c = 1$ convention may be written

$$\tilde{\pi}^{ij} = -\frac{3}{32\pi r^2} \left[p_\infty^i \hat{x}^j + p_\infty^j \hat{x}^i - (\delta^{ij} - \hat{x}^i \hat{x}^j)(p_\infty \cdot \hat{x}) \right], \quad (40)$$

which is trace free and divergence free away from the puncture [23, 24]. The vacuum Hamiltonian constraint becomes

$$\Delta_\delta \psi = -\frac{\kappa^2}{2} \psi^{-7} \tilde{\pi}^{ij} \tilde{\pi}_{ij}, \quad \psi = 1 + \frac{\sigma_0}{2r} + O_2(r^{-2}). \quad (41)$$

The seed indices in (41) are lowered with δ_{ij} . A puncture or an additional end requires its usual inner regularity and boundary conditions. This outer-end illustration is not a globally regular, one-ended vacuum solution on $\mathbb{R}^3$.

On the initial slice the spatial response can be represented by $e|_{T\Sigma} = \psi^2 \bar{e}|_{T\Sigma}$, whose leading scale coefficient is σ_0 . The normal deformation also contains the trace-free seed. This statement concerns the spatial soldering on one slice; conformal flatness of that slice does not make the full spacetime conformal to Minkowski spacetime. The resulting translation charges are

$$E_\infty^{\text{MS}} = \frac{8\pi\sigma_0}{\kappa}, \quad J_i^{\text{MS}} = -p_{\infty i}. \quad (42)$$

The momentum equality follows from $\int_{\mathbb{S}^2} \hat{x}_i \hat{x}_j d\Omega = (4\pi/3) \delta_{ij}$ and the sign $p_{\infty i} = -J_i^{\text{MS}}$. For $p_\infty = 0$ and $\sigma_0 > 0$, $\psi = 1 + \sigma_0/(2r)$ gives the time-symmetric Schwarzschild outer end. With nonzero momentum, the inner data and the elliptic solution determine σ_0 ; energy is not independently assigned after the seed has been chosen. The example is standard conformal initial-data theory, not an MS solution algorithm. Its role here is to show that the response scale and the trace-free normal deformation are coupled by the constraint: energy and momentum are not independent coefficients of an unconstrained response rescaling.

7.3 Physical units and ADM realization

Restore c . The boundary functional $\mathcal{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$ is energy-valued. We write

$$\mathcal{P}_\infty^{\text{MS}} = E_\infty^{\text{MS}} e^0 - c P_{\infty i}^{\text{MS}} e^i, \quad J_i^{\text{MS}} = -c P_{\infty i}^{\text{MS}}. \quad (43)$$

The associated momentum-valued positive-response covector is $\hat{\Pi}_\infty^{\text{MS}} = c^{-1} \mathcal{P}_\infty^{\text{MS}}$, and its Lorentz-polar vector is

$$P_\infty^{\text{MS}} = \frac{E_\infty^{\text{MS}}}{c} e_0 + P_{\infty i}^{\text{MS}} e_i. \quad (44)$$

The restricted classical completion fixes the relative form of the canonical generator but not the numerical coupling. We now impose the empirical Einstein normalization $\kappa = 8\pi G/c^4$. Equations (18) and (20) become

$$E_\infty^{\text{MS}} = \frac{c^4}{16\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_h, \quad (45)$$

$$P_{\infty i}^{\text{MS}} = -\frac{c^3}{8\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j{}_i - K \delta^j{}_i) s_j dS_h. \quad (46)$$

With the common convention $K_{ij}^{\text{ADM}} = -K_{ij}$, (46) is the standard ADM momentum formula.

Theorem 7.2 (ADM realization). *On the calibrated asymptotically Minkowskian phase space, in the torsion-free Einstein branch and after $\kappa = 8\pi G/c^4$ is imposed,*

$$E_\infty^{\text{MS}} = E_{\text{ADM}}, \quad \mathbf{P}_\infty^{\text{MS}} = \mathbf{P}_{\text{ADM}}, \quad P_\infty^{\text{MS}} = P_{\text{ADM}}. \quad (47)$$

The theorem is an identification of the MS boundary completion with the standard ADM translation charge, not a new canonical derivation of the ADM surface functional. Its content is the identification of the metric, canonical data and boundary term at each stage of the response reconstruction, with the Einstein dynamics and normalization explicitly supplied.

If the four-momentum is future timelike, define

$$M_\infty^{\text{MS}} = \frac{1}{c} \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})} = \frac{1}{c^2} \sqrt{(E_\infty^{\text{MS}})^2 - c^2 |\mathbf{P}_\infty^{\text{MS}}|^2}. \quad (48)$$

Then [theorem 7.2](#) gives $M_\infty^{\text{MS}} = M_{\text{ADM}}$, and the observer-energy formula reads

$$E_{\text{obs}}(D, S) = \frac{E_{\text{ADM}} - c \mathbf{P}_{\text{ADM}} \cdot D}{S}, \quad M_{\text{ADM}} c^2 = \min E_{\text{obs}} \quad (49)$$

when the ADM vector is future timelike. The minimizing coordinates are $D_\infty = c \mathbf{P}_{\text{ADM}} / E_{\text{ADM}}$ and $S_\infty = M_{\text{ADM}} c^2 / E_{\text{ADM}}$.

The causal classification in [\(32\)](#) must still be kept separate from positive mass. Under the standard hypotheses of the spacetime positive-mass theorem, $E_{\text{ADM}} \geq c |\mathbf{P}_{\text{ADM}}|$ and the total vector is future causal [\[9–11\]](#). Rigidity results exclude a nonzero null ADM vector under their corresponding regularity and decay assumptions [\[25–27\]](#). The equality conclusion is an embedding of the initial data into Minkowski spacetime; it need not mean $K_{ij} = 0$ or an intrinsically Euclidean metric in the chosen slicing. Inner boundaries require the hypotheses of the particular positive-mass theorem invoked. None of these conditions follows from response calibration alone.

8 Matter sources and stationary boundary charges

The ADM construction above uses the metric reconstructed from the local response but requires the Einstein dynamics as an additional input. The same separation should be kept when matter and an inner stationary boundary are present. In this section we set $c = 1$, retain $\hbar$ and k_B , take $\kappa = 8\pi G$, and assume zero cosmological constant with minimally coupled matter,

$$I[g, \text{matter}] = \frac{1}{2\kappa} \int_M R \epsilon_g + I_m[g, \text{matter}]. \quad (50)$$

The action and its coefficient are dynamical inputs. Deriving gravitational Hamiltonians and horizon terms from a common Einstein action has established precedents [\[5, 28\]](#); here the metric entering that action is supplied by the calibrated MS construction. The primitive response fixes the calibrated metric on which this action is evaluated; it does not by itself select the Einstein action or determine G . Matter fields are assumed to have sufficient falloff at the chosen outer end that they add no translation surface term. Additional gauge charges require their own boundary conditions.

8.1 The proper-time response as a matter source

For a classical massive particle let $U_A = dx_A/d\tau_A$, $g(U_A, U_A) = -1$, and $Q_A = m_A > 0$ in the present units. On the matched mass shell, the local response along the worldline gives $\tilde{\Theta}(\Pi_A, U_A) d\tau_A = -Q_A d\tau_A$. The corresponding generally covariant particle action is $I_A = -Q_A \int d\tau_A$. If an observer resolves the same proper-time increment by $S_A = d\tau_A/d\tau_{u,A}$

and $q_A = Q_A S_A$, then $Q_A d\tau_A = q_A d\tau_{u,A}$. This is an observer-dependent readout of one invariant action, not a replacement of Q_A by a different source mass.

For the foliation observer n , the assigned local time increment is $d\tau_{n,A} = N dt$. The same action has coordinate Lagrangian $L_A = -N q_A$, whereas its Hamiltonian is $H_A = N Q_A / S_A - \beta^i p_{Ai}$. Thus q_A is an action readout; Q_A / S_A is the observer energy. This comparison applies to momentum matched to the trajectory, not to arbitrarily fixed independent response labels.

Metric variation of the minimally coupled proper-time action at fixed worldlines gives the standard distributional stress tensor [29]

$$T^{\mu\nu}(x) = \sum_A Q_A \int U_A^\mu U_A^\nu \frac{\delta^{(4)}(x - x_A(\tau_A))}{\sqrt{-g(x)}} d\tau_A. \quad (51)$$

The distributional notation specifies the source normalization in a test-particle or regulated matter treatment; it does not establish an exact nonlinear point-source Einstein solution. The same invariant scale Q_A normalizes the proper-time response and the minimally coupled matter source. The total ADM mass is not $\sum_A Q_A$: kinetic energy, field energy and gravitational binding belong to the global solution. The bridge in (51) comes from minimal coupling to the calibrated metric and the Einstein–matter action, not from a new derivation of stress–energy from kinematics.

8.2 One covariant boundary variation

Let ϕ denote all metric and matter fields and write the Lagrangian four-form as L . In the covariant phase-space formalism [30, 31],

$$\delta L = \mathcal{E} \cdot \delta\phi + d\vartheta(\phi, \delta\phi), \quad J_\xi = \vartheta(\phi, \mathcal{L}_\xi \phi) - \iota_\xi L. \quad (52)$$

On shell $J_\xi = dQ_\xi$ locally. The presymplectic potential ϑ is a property of the field action and should not be confused with the primitive scalar response $\tilde{\Theta}$. For a fixed vector field ξ and variations tangent to the solution space, the boundary one-form defines a Hamiltonian variation when it is integrable under the selected boundary conditions:

$$\delta H_\xi[S] = \int_S (\delta Q_\xi - \iota_\xi \vartheta), \quad Q_\xi = -\frac{1}{2\kappa} \star d\xi^\flat, \quad (53)$$

Here S is a boundary two-surface; we use the Einstein representative and a consistent orientation. For a background symmetry $\mathcal{L}_\xi \phi = 0$ and variations satisfying the linearized field equations, the two-form $\delta Q_\xi - \iota_\xi \vartheta$ is closed. Its integrals can then be compared at infinity and at a stationary inner boundary. Away from this setting, presymplectic flux and integrability require separate treatment. This common boundary framework is established in covariant phase space and covariant Hamiltonian gravity [5, 30]. At spatial infinity, the constant-translation specialization agrees with the ADM Hamiltonians identified in [theorem 7.2](#). Rotations require the standard Regge–Teitelboim extension with their additional parity and growth conditions [1, 3]; that extension is not derived from the translation-sector argument above.

For the stationary comparison, let k be the unit time translation at infinity and m an axial Killing field of period 2π . With the usual Hamiltonian convention, $H_k = M$ and $H_m = -J$, where $M = M_{\text{ADM}}$ in the stationary rest frame. The symbol J here denotes angular momentum and is unrelated to the spatial translation coefficients J_i^{MS} .

8.3 Horizon boost charge and the first law

Consider a nonextremal, stationary, asymptotically flat vacuum black hole with bifurcation surface B and horizon generator $\xi_H = k + \Omega_H m$. In the fixed-generator variation, ξ_H is held fixed.

Since it vanishes on B , the covariant boundary identity gives the standard first law [30–32],

$$\delta M - \Omega_H \delta J = \frac{\kappa_H}{\kappa} \delta A_H. \quad (54)$$

The area/normal-plane-angle relation has prior formulations in black-hole action and boundary phase space [33]; the following boost interpretation is an Einstein–Noether realization of that established structure. The vector field ξ_H/κ_H generates a unit boost in the horizon normal plane. Its oriented Einstein Noether charge is $P_\eta := \int_B Q_{\xi_H/\kappa_H} = A_H/\kappa$. The associated Wald entropy is therefore

$$S_H = \frac{2\pi k_B}{\hbar} P_\eta = \frac{k_B A_H}{4\hbar G} \quad (55)$$

with $c = 1$; restoring physical units supplies the usual factor c^3 in the numerator. Hawking’s quantum-field-theoretic temperature $T_H = \hbar\kappa_H/(2\pi k_B)$ then turns (54) into $\delta M - \Omega_H \delta J = T_H \delta S_H$ [34]. Neither T_H nor a microscopic state count is obtained from the response pairing.

8.4 Komar normalization and the remaining coupling

In the stationary vacuum exterior the Komar flux can be moved from infinity to the horizon. With the mass normalization used here,

$$K[\xi_H] := -\frac{1}{\kappa} \int_S \star d\xi_H^\flat = M - 2\Omega_H J = \frac{2\kappa_H A_H}{\kappa}, \quad (56)$$

which is the vacuum Smarr relation [35, 36]. The Hamiltonian variation contains $\delta M - \Omega_H \delta J$, not the variation of $M - 2\Omega_H J$ with all background coefficients varied. The presymplectic term in (53) and the different normalizations of mass and angular momentum are essential to this distinction.

Equations (51), (47) and the horizon boost charge place the local proper-time source, the asymptotic Hamiltonian charge and the stationary horizon charge in one calibrated metric and one variational theory. The asymptotic and horizon gravitational charges share the factor κ^{-1} , whereas the particle source carries its invariant matter scale Q_A . The equation $G_{\mu\nu} = \kappa T_{\mu\nu}$ compares these two normalizations. Nothing in the response calibration fixes that relative coupling. Changing the coefficient of the Einstein action leaves the local soldered metric kinematics intact but changes the gravitational response to a fixed matter source; an explanation of G therefore requires additional dynamics or microscopic input.

9 Discussion and outlook

The asymptotic part of the paper identifies the total translation charge of the classical MS sector. The local response reconstructs a soldered Lorentzian geometry; the separately chosen Einstein branch supplies the bulk constraints; and allowing lapse and shift to approach nonzero translations exposes the standard differentiability obstruction. Its boundary primitive is linear in the asymptotic translation parameter and therefore defines $\mathcal{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$. After the empirical normalization $\kappa = 8\pi G/c^4$, this covector is the ADM energy–momentum.

Three distinctions remain useful throughout. Internal Lorentz gauge changes the soldering representative but not the metric. Response recalibration changes the metric scale and can change the asymptotic data. The coefficient κ , by contrast, assigns a gravitational charge to a fixed canonical geometry. The response-jet calculation makes the first distinction concrete: the translation surface functionals see only the monopole of the leading spatial scale and the dipole of its normal first jet, whereas the constraints determine which such jets occur in actual initial data. The conformal example shows explicitly that the leading scale and trace-free normal deformation

are coupled by the elliptic constraint rather than freely specifiable as four independent response modes.

The observer-space description packages the completed charge in the same compact variables used locally. In the future-timelike sector, the invariant mass is the minimum observer energy, while the first hyperbolic derivative reconstructs the full total vector. The Hermitian representation in [section 6.2](#) adds only an algebraic interface: once future causality is known, the normalized charge lies in the trace-one convex section of the positive Hermitian cone and its determinant reproduces the global staticity. The cone correspondence and its rank-one boundary are established algebraic geometry [\[19\]](#); the present paper does not use that representation to assume positive mass.

The stationary section extends the same classical architecture rather than introducing a second gravitational normalization. Variation of the matter action with respect to the calibrated metric gives the standard particle stress tensor, while the diffeomorphism boundary variation gives the ADM Hamiltonians at infinity and, under stationary vacuum hypotheses, the horizon boost charge. The first law, Wald area charge and Komar–Smarr relation retain their standard status. What the MS formulation supplies is a common response-derived metric and a clear accounting of which normalization enters each charge. In particular, neither the local proper-time scale nor the equality of boundary variations determines Newton’s constant.

Positive energy remains a further global theorem. The Hamiltonian construction classifies any completed translation vector but does not force it into the future cone. Under the usual completeness, decay and dominant-energy assumptions one may invoke the standard spacetime positive-mass theorem; a companion analysis rewrites the corresponding directional quantities as null MS responses and studies their Sen–Witten frame realization [\[12\]](#). The separation is deliberate: charge construction, cone positivity and spinorial gauge selection are different logical steps.

The next classical extensions are correspondingly specific. A full asymptotic Poincaré completion should add differentiable boosts, rotations and center-of-mass charges; enlarged asymptotic symmetry classes should determine which higher response modes survive beyond the translation quotient; and nonstationary inner boundaries require control of presymplectic flux rather than the stationary identities used here. On the quantum side, the local soldering Weyl algebra of [Ref. \[8\]](#) remains a kinematical comparison algebra. A positive physical boundary Hamiltonian requires a constrained Hilbert space, a compatible ordering and genuine asymptotic generators; it cannot be obtained by simply identifying an unconstrained additive response operator with ADM energy.

The resulting statement is a conditional realization within Einstein gravity. The local response supplies the calibrated metric, the chosen field action supplies the bulk generator, and its asymptotic differentiability supplies the translation charge. Stationary boundary variation then relates that charge to the horizon boost term. The Hamiltonian and Noether mechanisms are established; the MS-specific content is the common response parametrization and the explicit separation of calibration, constraints and gravitational coupling.

A Finite-radius boundary variation

For completeness we record the boundary terms used in [section 4](#). Let

$$V_R[N] = -\frac{1}{2\kappa} \int_{\Sigma_R} N \sqrt{h} {}^{(3)}R d^3x. \quad (57)$$

For covariant metric variations, put $\delta h = h^{ij} \delta h_{ij}$. The standard identity is

$$\delta(\sqrt{h}R) = \sqrt{h} \left[-G^{ij} \delta h_{ij} + D^i D^j \delta h_{ij} - D^2 \delta h \right]. \quad (58)$$

Integrating the derivative terms by parts at fixed N gives the boundary term

$$\mathcal{S}_R^\perp[N; \delta h] = -\frac{1}{2\kappa} \oint_{S_R} s^i \left[N(D^j \delta h_{ij} - D_i \delta h) - (D^j N) \delta h_{ij} + (D_i N) \delta h \right] dS_h, \quad (59)$$

which is (17). The terms involving $D_i N$ vanish for translations because $D_i N = O(r^{-2})$ and $\delta h_{ij} = O(r^{-1})$. Replacing physical by reference normals and derivatives changes the integrand only by $o(r^{-2})$, so the surviving limit is $-a^\perp \delta E_\infty^{\text{MS}}$.

For the tangential part, because π^{ij} has density weight $+1$,

$$-2 \int_{\Sigma_R} \beta^i D_j \pi^j_i d^3x = \int_{\Sigma_R} \pi^{ij} \mathcal{L}_\beta h_{ij} d^3x - 2 \oint_{S_R} \beta^i \pi_i^j dS_j. \quad (60)$$

Varying the bulk term and integrating derivatives of δh_{ij} back into the interior leaves, besides the variation of the density flux, the finite-radius remainder

$$\mathcal{R}_R^\parallel[\beta; \delta h] = \oint_{S_R} \beta^k \pi^{ij} \delta h_{ij} dS_k. \quad (61)$$

The asymptotic orders $\beta^k = O(1)$, $\pi^{ij} = O(r^{-2})$, $\delta h_{ij} = O(r^{-1})$ make this $O(R^{-1})$. Hence the only finite variation is $-a^i \delta J_i^{\text{MS}}$.

B Asymptotic translation estimates and conventions

For a constant asymptotic translation, Hamilton's equation for the spatial metric has the schematic form

$$\delta_a h_{ij} = \frac{4\kappa N_a}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) + \mathcal{L}_{\beta_a} h_{ij}. \quad (62)$$

The falloff $h_{ij} = \delta_{ij} + O_2(r^{-1})$, $\pi^{ij} = O_1(r^{-2})$ and constant leading N_a, β_a^i imply $\delta_a h_{ij} = O(r^{-2})$. The momentum equation similarly gives $\delta_a \pi^{ij} = O(r^{-3})$: its curvature term has that order, the quadratic momentum terms are $O(r^{-4})$, and the constant asymptotic shift differentiates the leading r^{-2} density. For representatives exactly constant near infinity, lapse-derivative terms vanish there. These estimates are sufficient for the invariance of the leading translation charges used in [theorem 5.1](#).

Our sign convention for extrinsic curvature is $K_{ij} = (2N)^{-1}(\dot{h}_{ij} - \mathcal{L}_\beta h_{ij})$. If $K_{ij}^{\text{ADM}} = -K_{ij}$ is used, then the physical momentum formula becomes

$$P_{\text{ADM}i} = \frac{c^3}{8\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (K^{\text{ADM}j}_i - K^{\text{ADM}} \delta^j_i) s_j dS_h. \quad (63)$$

The density surface element dS_j and geometric element satisfy $\sqrt{h} dS_j = s_j dS_h$.

C Observer-space reconstruction and calibration quotients

C.1 Intrinsic characterization of linear observer energies

Let E be a smooth function on H^3 satisfying $\text{Hess}_H E = E h_H$. Fix $u_0 \in H^3$ and define an ambient vector by $P = E(u_0)u_0 - \nabla_H E|_{u_0}$. Along any geodesic through u_0 , the Hessian equation reduces to $E'' = E$, so the initial value and first derivative uniquely determine E . It follows that $E(u) = -\eta(P, u)$ everywhere. Thus the Hessian equation used in [proposition 6.1](#) characterizes restrictions of ambient Lorentzian linear functionals. This four-dimensional solution space is the standard hyperbolic static-potential space [[15](#), [16](#)], not an additional MS reconstruction theorem.

The first integral $E^2 - |\nabla_H E|^2$ is constant because

$$d(E^2 - |\nabla_H E|^2) = 2E dE - 2 \text{Hess}_H E(\nabla_H E, \cdot) = 0. \quad (64)$$

Its sign distinguishes the timelike and spacelike cases; a zero value permits a null or zero vector. Time orientation, and the distinction between null and zero, are read from the energy function itself.

C.2 Translation-visible calibration quotient

At the leading translation-sensitive orders, define the response-jet space

$$\mathcal{J}_{\text{tr}} = C_{\text{even}}^\infty(\mathbb{S}^2) \oplus C_{\text{odd}}^\infty(\mathbb{S}^2) \quad (65)$$

and the readout map

$$\mathcal{C}_{\text{tr}}(\sigma, \rho) = \left(\int_{\mathbb{S}^2} \sigma d\Omega, \int_{\mathbb{S}^2} \rho \hat{x}_i d\Omega \right). \quad (66)$$

It is surjective: constants generate the scalar component, while $\rho = v_i \hat{x}^i$ generates an arbitrary vector because $\int_{\mathbb{S}^2} \hat{x}_i \hat{x}_j d\Omega = (4\pi/3) \delta_{ij}$. Hence

$$\mathcal{J}_{\text{tr}} / \ker \mathcal{C}_{\text{tr}} \simeq \mathbb{R} \oplus \mathbb{R}^3, \quad (67)$$

corresponding to the $\ell = 0$ part of σ and $\ell = 1$ part of ρ . This quotient is kinematic and follows from elementary spherical-harmonic projection. The strong calibration (13) sets the entire leading jet to zero. Requiring only that the four integrated translation charges remain fixed is weaker: at the displayed orders it requires $\int \sigma d\Omega = 0$ and $\int \rho \hat{x}_i d\Omega = 0$. Such charge preservation neither makes the transformation gauge nor guarantees constraint admissibility. Proposition 7.1 shows how restricted the pure-rescaling family becomes after the constraints are imposed.

Subleading changes of a calibrated flat reference are also invisible to the energy if $\bar{h}'_{ij} - \bar{h}_{ij} = o(r^{-1})$ with differentiated remainder $o(r^{-2})$: their surface contribution tends to zero. The spatial charge is defined from the physical K or π , with the reference slice fixed by $\bar{K} = 0$; it has no independent reference- K subtraction in the prescription used here.

References

- [1] Tullio Regge and Claudio Teitelboim. Role of surface integrals in the hamiltonian formulation of general relativity. *Annals of Physics*, 88(1):286–318, 1974. doi: 10.1016/0003-4916(74)90404-7.
- [2] Richard Arnowitt, Stanley Deser, and Charles W. Misner. The dynamics of general relativity. In Louis Witten, editor, *Gravitation: An Introduction to Current Research*, pages 227–264. John Wiley & Sons, New York, 1962.
- [3] Robert Beig and Niall Ó Murchadha. The poincaré group as the symmetry group of canonical general relativity. *Annals of Physics*, 174(2):463–498, 1987. doi: 10.1016/0003-4916(87)90037-6.
- [4] J. David Brown and James W. York, Jr. Quasilocal energy and conserved charges derived from the gravitational action. *Physical Review D*, 47(4):1407–1419, 1993. doi: 10.1103/PhysRevD.47.1407.
- [5] Chiang-Mei Chen and James M. Nester. Quasilocal quantities for general relativity and other gravity theories. *Classical and Quantum Gravity*, 16(4):1279–1304, 1999. doi: 10.1088/0264-9381/16/4/018.

- [6] Mingyang Han. Decomposition rigidity and compact-state duality in relativistic free-particle dynamics. Unpublished manuscript, 2026.
- [7] Mingyang Han. Operational soldering from relativistic massive-state geometry: An inverse reconstruction. Unpublished manuscript, 2026.
- [8] Mingyang Han. From operational soldering to quantum configuration space: Coherent deformations and classical concentration. Unpublished manuscript, 2026.
- [9] Richard Schoen and Shing-Tung Yau. On the proof of the positive mass conjecture in general relativity. *Communications in Mathematical Physics*, 65(1):45–76, 1979. doi: 10.1007/BF01940959.
- [10] Edward Witten. A new proof of the positive energy theorem. *Communications in Mathematical Physics*, 80(3):381–402, 1981. doi: 10.1007/BF01208277.
- [11] Michael Eichmair, Lan-Hsuan Huang, Dan A. Lee, and Richard Schoen. The spacetime positive mass theorem in dimensions less than eight. *Journal of the European Mathematical Society*, 18(1):83–121, 2016. doi: 10.4171/JEMS/584.
- [12] Mingyang Han. MS null responses and positive energy: Classical frame identities, gauge selection and quantum models. Unpublished research note, 2026.
- [13] Ettore Minguzzi. Lorentzian causality theory. *Living Reviews in Relativity*, 22(1):3, 2019. doi: 10.1007/s41114-019-0019-x.
- [14] Abhay Ashtekar and R. O. Hansen. A unified treatment of null and spatial infinity in general relativity. i. universal structure, asymptotic symmetries, and conserved quantities at spatial infinity. *Journal of Mathematical Physics*, 19:1542–1566, 1978. doi: 10.1063/1.523863.
- [15] Piotr T. Chruściel and Marc Herzlich. The mass of asymptotically hyperbolic riemannian manifolds. *Pacific Journal of Mathematics*, 212(2):231–264, 2003. doi: 10.2140/pjm.2003.212.231.
- [16] Lan-Hsuan Huang, Hyun Chul Jang, and Daniel Martin. Mass rigidity for hyperbolic manifolds. *Communications in Mathematical Physics*, 376(3):2329–2349, 2020. doi: 10.1007/s00220-019-03623-0.
- [17] John G. Ratcliffe. *Foundations of Hyperbolic Manifolds*, volume 149 of *Graduate Texts in Mathematics*. Springer, Cham, 3 edition, 2019. doi: 10.1007/978-3-030-31597-9.
- [18] Po-Ning Chen, Pei-Ken Hung, Mu-Tao Wang, and Shing-Tung Yau. The rest mass of an asymptotically anti-de Sitter spacetime. *Annales Henri Poincaré*, 18(5):1493–1518, 2017. doi: 10.1007/s00023-017-0555-4.
- [19] Pablo Arrighi and Christophe Patricot. A note on the correspondence between qubit quantum operations and special relativity. *Journal of Physics A: Mathematical and General*, 36(20):L287–L296, 2003. doi: 10.1088/0305-4470/36/20/101.
- [20] Walter Simon. Conformal positive mass theorems. *Letters in Mathematical Physics*, 50:275–281, 1999. doi: 10.1023/A:1007637205709. URL <https://arxiv.org/abs/gr-qc/0003106>.
- [21] André Lichnerowicz. L’intégration des équations de la gravitation relativiste et le problème des n corps. *Journal de Mathématiques Pures et Appliquées*, 23:37–63, 1944. URL https://www.numdam.org/item/JMPA_1944_9_23__37_0/. Série 9.
- [22] James W. York, Jr. Gravitational degrees of freedom and the initial-value problem. *Physical Review Letters*, 26(26):1656–1658, 1971. doi: 10.1103/PhysRevLett.26.1656.

- [23] Jeffrey M. Bowen and James W. York, Jr. Time-asymmetric initial data for black holes and black-hole collisions. *Physical Review D*, 21(8):2047–2056, 1980. doi: 10.1103/PhysRevD.21.2047.
- [24] Steven Brandt and Bernd Brügmann. A simple construction of initial data for multiple black holes. *Physical Review Letters*, 78(19):3606–3609, 1997. doi: 10.1103/PhysRevLett.78.3606.
- [25] Abhay Ashtekar and Gary T. Horowitz. Energy-momentum of isolated systems cannot be null. *Physics Letters A*, 89(4):181–184, 1982. doi: 10.1016/0375-9601(82)90852-7.
- [26] Lan-Hsuan Huang and Dan A. Lee. Equality in the spacetime positive mass theorem. *Communications in Mathematical Physics*, 376(3):2379–2407, 2020. doi: 10.1007/s00220-019-03619-w.
- [27] Lan-Hsuan Huang and Dan A. Lee. Equality in the spacetime positive mass theorem ii. *Calculus of Variations and Partial Differential Equations*, 64(3):92, 2025. doi: 10.1007/s00526-025-02947-1.
- [28] Stephen W. Hawking and Gary T. Horowitz. The gravitational hamiltonian, action, entropy and surface terms. *Classical and Quantum Gravity*, 13(6):1487–1498, 1996. doi: 10.1088/0264-9381/13/6/017.
- [29] Robert M. Wald. *General Relativity*. University of Chicago Press, Chicago, 1984. ISBN 9780226870335. URL <https://press.uchicago.edu/ucp/books/book/chicago/G/bo5952261.html>.
- [30] Vivek Iyer and Robert M. Wald. Some properties of noether charge and a proposal for dynamical black hole entropy. *Physical Review D*, 50(2):846–864, 1994. doi: 10.1103/PhysRevD.50.846.
- [31] Robert M. Wald. Black hole entropy is the noether charge. *Physical Review D*, 48(8):R3427–R3431, 1993. doi: 10.1103/PhysRevD.48.R3427.
- [32] James M. Bardeen, Brandon Carter, and Stephen W. Hawking. The four laws of black hole mechanics. *Communications in Mathematical Physics*, 31:161–170, 1973. doi: 10.1007/BF01645742.
- [33] Steven Carlip and Claudio Teitelboim. The off-shell black hole. *Classical and Quantum Gravity*, 12(7):1699–1704, 1995. doi: 10.1088/0264-9381/12/7/011.
- [34] Stephen W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975. doi: 10.1007/BF02345020. Erratum: *Communications in Mathematical Physics* 46, 206 (1976).
- [35] Arthur Komar. Covariant conservation laws in general relativity. *Physical Review*, 113(3):934–935, 1959. doi: 10.1103/PhysRev.113.934.
- [36] Larry Smarr. Mass formula for kerr black holes. *Physical Review Letters*, 30:71–73, 1973. doi: 10.1103/PhysRevLett.30.71. Erratum: *Physical Review Letters* 30, 521 (1973).